Lab 02-Greedy Algorithm

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∗ Contact TA Yixuan Cai for any questions. Include both your .pdf and .tex files in the uploaded .rar or .zip file.

∗ Name:Hengtao Wu Student ID:524031910038 Email: Eternity_w@sjtu.edu.cn

1. (Fractional Knapsack) Let $(w_1, v_1), (w_2, v_2), \ldots, (w_n, v_n)$ be n pairs of positive real values.

   (a) Given a real value $W \le \sum_{i=1}^{n} w_i$, design an algorithm to find $x_1, x_2, \ldots, x_n$ to maximize the objective function
   $$\sum_{i=1}^{n} \frac{v_i}{w_i} \cdot x_i$$
   subject to
   - $0 \le x_i \le w_i$ for every $i \in [1, n]$;
   - $\sum_{i=1}^{n} x_i \le W$.

   (b) Analyze the time complexity of your algorithm.

Solution. (a) **Algorithm Principle:** The algorithm greedily selects items with the highest ratio (v_i/w_i), taking them fully if capacity allows, or fractionally to fill the knapsack. The pseudo-code is presented below.

Algorithm 1: Fractional Knapsack Greedy Algorithm

Input : $(w_1, v_1), (w_2, v_2), \ldots, (w_n, v_n)$, W
Output: $A = \{x_1, x_2, \ldots, x_n\}$

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Initialize x_i = 0 for every i ∈ [1, n];
Sort the items such that v_1/w_1 ≥ v_2/w_2 ≥ ... ≥ v_n/w_n;
for i = 1, 2, ..., n do
    if W == 0 then
        break;
    if W ≥ w_i then
        x_i ← w_i ;                                  // Greedy
        W ← W − w_i;
    else
        x_i ← W;
        W ← 0;
        break;
return A = {x_1, x_2, ..., x_n};
```

Proof of Optimality (Exchange Argument):

Let the items be sorted by their value-to-weight ratio such that $\frac{v_1}{w_1} \ge \frac{v_2}{w_2} \ge \ldots \ge \frac{v_n}{w_n}$. Let $G = \{x_1, x_2, \ldots, x_n\}$ be the solution produced by our greedy algorithm, and $O = \{y_1, y_2, \ldots, y_n\}$ be an optimal solution. Assume, for the sake of contradiction, that $G \ne O$.

Let i be the first index where the two solutions differ (i.e., $x_k = y_k$ for all $k < i$, and $x_i \ne y_i$). Because the greedy algorithm always takes as much of the highest-ratio available item as possible, we must have $x_i > y_i$. To compensate for taking less of item i, the optimal solution O must have taken more of some other item j (where $j > i$) compared to the greedy solution. Since the items are sorted, we know $\frac{v_i}{w_i} \ge \frac{v_j}{w_j}$.

Consider a small weight $\Delta = \min(x_i - y_i, y_j) > 0$. If we remove Δ weight of item j from O and replace it with Δ weight of item i, the total weight of the knapsack remains valid and unchanged. However, the total value of this new solution changes by:

$$\Delta \cdot \left(\frac{v_i}{w_i} - \frac{v_j}{w_j} \right) \geq 0$$

This exchange yields a new valid solution that is at least as valuable as O. By repeatedly applying this exchange process for all differences, we can eventually transform the optimal solution O entirely into the greedy solution G without ever decreasing the total value. Therefore, the greedy solution G must also be a optimum.

(b) Complexity Analysis

- **Sorting:** Calculating the fraction $\frac{v_i}{w_i}$ for each of the n items takes $\mathcal{O}(n)$ time. Sorting these n items in descending order based on their fractions takes $\mathcal{O}(n \log n)$ time.
- **Greedy Choice:** The algorithm iterates through the sorted items using a single **for** loop. In the worst case, it runs n times. Inside the loop, all conditional checks take constant time $\mathcal{O}(1)$. Thus, this greedy selection phase takes $\mathcal{O}(n)$ time.

Overall,the time complexity is dominated by the sorting step, which yields a total time complexity of $\mathcal{O}(n \log n) + \mathcal{O}(n) = \mathcal{O}(n \log n)$.

□

2. (Min Interval Covering) Given a set of n distinct points $\{x_1, x_2, \ldots, x_n\}$ distributed on real number axis with $x_1 < x_2 < \cdots < x_n$, cover these points using closed intervals of length 1.

   (a) Design an algorithm to find the minimum number of such intervals and their positions.

   (b) Provide a correctness proof of your algorithm.

Solution. (a) Algorithm design. The algorithm iterates through the points from left to right. Whenever it encounters the first uncovered point x_i, it places an interval starting exactly at x_i, extending to $x_i + 1$. This greedily covers as many subsequent points as possible before needing a new interval.

Algorithm 2: Greedy Min Interval Covering

Input : A set of distinct points $\{x_1, x_2, \ldots, x_n\}$ with $x_1 < x_2 < \ldots < x_n$
Output: Minimum number of intervals *count*, and their positions I

Initialize $I = \emptyset$ $count = 0$ $i = 1$;
while $i \leq n$ **do**
    Add the interval $[x_i, x_i + 1]$ to I;
    $count \leftarrow count + 1$;
    `// Find the first point not covered by` $[x_i, x_i + 1]$
    Let x_k be the smallest point such that $x_k > x_i + 1$;
    if *such* x_k *exists* **then**
        $i \leftarrow k$;
    else
        break;
return *count*, and I;

(b) **Theorem.**Greedy Min Interval Covering algorithm is optimal.
Proof. Let $G = \{g_1, g_2, \ldots, g_k\}$ be the sequence of intervals chosen by our greedy algorithm, and $O = \{o_1, o_2, \ldots, o_m\}$ be the intervals of an optimal solution.

Assume, for the sake of contradiction, that $G \neq O$. Let j be the first index where the two solutions differ (i.e., $g_r = o_r$ for all $r < j$, but $g_j \neq o_j$).

Let x^* be the first point not covered by the first $j-1$ intervals. By our greedy strategy, the algorithm places $g_j = [x^*, x^* + 1]$.Since the optimal solution O is valid, it must also cover x^*. Therefore, its j-th interval must be $o_j = [y, y+1]$ where $y \leq x^*$. Because $y \leq x^*$, it strictly implies that the right endpoint $y + 1 \leq x^* + 1$.

This means the right endpoint of the greedy interval g_j extends at least as far to the right as the optimal interval o_j. Hence, g_j covers all the unhandled points that o_j covers.

If we exchange o_j with g_j in the optimal solution O, no points will be left uncovered.This exchange creates a new valid solution that uses the exact same number of intervals as O (thus still optimal), but matches G up to the j-th interval.

By inductively repeating this exchange process for all differences, we can transform the optimal solution O entirely into the greedy solution G without ever increasing the number of intervals. Therefore, the greedy algorithm produces a minimum set of intervals.

□

3. (Matroid) Consider a pair $M = (U, \mathcal{I})$ where universe U is a finite set and $\mathcal{I} \subseteq 2^U$ is a collection of subsets of U. We call M a matroid if it satisfies *hereditary property* and *exchange property*:

- (*HEREDITARY PROPERTY*) $\mathcal{I}$ is nonempty and for every $A \in \mathcal{I}$ and $B \subseteq A$, it holds that $B \in \mathcal{I}$.
- (*EXCHANGE PROPERTY*) For any $A, B \in \mathcal{I}$ with $|A| < |B|$, there exists some $x \in B \setminus A$ such that $A \cup \{x\} \in \mathcal{I}$.

Each set $A \in \mathcal{I}$ is called an *independent set*. The maximal independent sets are called *bases*.

(a) The dual of a matroid $\mathcal{I}$ over a universe U is the family of subsets $\mathcal{I}'$ over the same universe U, consisting of the complements of bases of $\mathcal{I}$, and all subsets of these sets. Prove that the dual $\mathcal{I}'$ is itself a matroid.

(b) Consider a matroid $\mathcal{I}$ over a universe U, and for a parameter k, define the following family of subsets

$$\mathcal{I}_k = \{A \mid A \in I \text{ and } |A| \leq k\}$$

Prove that for each positive integer k, $\mathcal{I}_k$ is a matroid.

Solution. (a) **Proof.** We demonstrate that the dual $\mathcal{I}'$ is a matroid by verifying the hereditary property and the exchange property.

- (*HEREDITARY PROPERTY*) Let $A \in \mathcal{I}'$ and $A' \subseteq A$. By the definition of the dual matroid, there exists a base B of $\mathcal{I}$ such that A is a subset of the complement of B, i.e., $A \subseteq U \setminus B$. Since $A' \subseteq A$, it follows directly that $A' \subseteq U \setminus B$. Thus, A' is also a subset of the complement of a base, meaning $A' \in \mathcal{I}'$.
- (*EXCHANGE PROPERTY*) Let $A, B \in \mathcal{I}'$ with $|A| < |B|$. We need to find an element $x \in B \setminus A$ such that $A \cup \{x\} \in \mathcal{I}'$.
Since $A, B \in \mathcal{I}'$, there exist bases B_A, B_B of $\mathcal{I}$ such that $B_A \cap A = \emptyset$ and $B_B \cap B = \emptyset$.

To prove $A \cup \{x\} \in \mathcal{I}'$, it suffices to construct a base $\hat{B}$ of $\mathcal{I}$ satisfying $\hat{B} \cap A = \emptyset$ and $\hat{B} \cap \{x\} = \emptyset$ for some $x \in B \setminus A$.
Let $B_B \cap A = \{e_1, e_2, \ldots, e_k\}$. We construct $\hat{B}$ from B_B by iteratively replacing each $e_i \in A$ with a safe element from B_A. Specifically, by the base exchange property of matroids, for each e_i, there exists $f_i \in B_A$ such that $(B_B \setminus \{e_i\}) \cup \{f_i\}$ is again a base. Applying this exchange k times yields a base $\hat{B}$ with $\hat{B} \cap A = \emptyset$, since all elements of $B_B \cap A$ have been removed and replaced by elements of $B_A \subseteq U \setminus A$.
It remains to show that $\hat{B}$ does not contain all elements of $B \setminus A$. Since $B_B \cap B = \emptyset$, the only elements of B that could appear in $\hat{B}$ are those introduced via the f_i's. Therefore:

$$|\hat{B} \cap (B \setminus A)| \leq k = |B_B \cap A|$$

Since $B_B \cap B = \emptyset$, every element of $B_B \cap A$ belongs to $A \setminus B$, so $k \leq |A \setminus B|$. Combined with the assumption $|A| < |B|$, which implies $|A \setminus B| < |B \setminus A|$, we obtain:

$$|\hat{B} \cap (B \setminus A)| \leq |A \setminus B| < |B \setminus A|$$

Hence $\hat{B}$ does not contain every element of $B \setminus A$. There exists at least one $x \in B \setminus A$ with $x \notin \hat{B}$. Since $\hat{B} \cap A = \emptyset$ and $x \notin \hat{B}$, we have $\hat{B} \subseteq U \setminus (A \cup \{x\})$, which gives $A \cup \{x\} \in \mathcal{I}'$.

$\square$

(b) **Proof.** We verify the two properties for $\mathcal{I}_k = \{A \mid A \in \mathcal{I} \text{ and } |A| \leq k\}$.

- (*HEREDITARY PROPERTY*) Let $A \in \mathcal{I}_k$ and $B \subseteq A$. Since $A \in \mathcal{I}_k$, we know $A \in \mathcal{I}$ and $|A| \leq k$. Because $\mathcal{I}$ is a matroid, the hereditary property guarantees that any subset of an independent set is independent, so $B \in \mathcal{I}$. Furthermore, since $B \subseteq A$, we have $|B| \leq |A| \leq k$. Thus, $B \in \mathcal{I}$ and $|B| \leq k$, meaning $B \in \mathcal{I}_k$.
- (*EXCHANGE PROPERTY*) Let $A, B \in \mathcal{I}_k$ with $|A| < |B|$. Since $\mathcal{I}_k \subseteq \mathcal{I}$, both A and B are independent sets in the original matroid $\mathcal{I}$. By the exchange property of $\mathcal{I}$, there exists an element $x \in B \setminus A$ such that $A \cup \{x\} \in \mathcal{I}$.
We only need to verify the size constraint for $A \cup \{x\}$. Since $|A| < |B|$ and both are integers, we have $|A| \leq |B| - 1$. We also know that $B \in \mathcal{I}_k$, which implies $|B| \leq k$. Substituting this gives $|A| \leq k - 1$. Therefore, the size of the new set is:

$$|A \cup \{x\}| = |A| + 1 \leq (k-1) + 1 = k$$

Since $A \cup \{x\} \in \mathcal{I}$ and $|A \cup \{x\}| \leq k$, we conclude that $A \cup \{x\} \in \mathcal{I}_k$.

$\square$

$\square$

Appendix: An Alternative Approach (For TA's Feedback)

Dear TA, the proof above is my final answer. I actually tried another method first (the "capacity" argument below), but I wasn't 100% sure about it. Could you help check if this alternative proof is also acceptable, or if I missed something subtle? Really appreciate your feedback!

Alternative Proof. We demonstrate the exchange property by building a new base and analyzing its capacity. Let $A, B \in \mathcal{I}'$ with $|A| < |B|$. We need to find an element $x \in B \setminus A$ such that $A \cup \{x\} \in \mathcal{I}'$.

Since $A, B \in \mathcal{I}'$, there exist bases B_A, B_B of $\mathcal{I}$ such that $A \subseteq U \setminus B_A$ and $B \subseteq U \setminus B_B$. Taking the complement of both sides gives $B_A \subseteq U \setminus A$ and $B_B \subseteq U \setminus B$.

Consider the set $I = B_B \setminus A$. Since $I \subseteq B_B \in \mathcal{I}$, I is an independent set in $\mathcal{I}$. Moreover, $I \subseteq U \setminus A$ by construction.

We claim we can extend I to a base B^* of $\mathcal{I}$ that is entirely contained in $U \setminus A$. To see this, note that $B_A \subseteq U \setminus A$ and B_A is a base of $\mathcal{I}$, so $r(U \setminus A) = r(U) = r$ (the rank of the whole matroid). Since $I \subseteq U \setminus A$ is independent, we can extend I to a maximal independent set B^* within $U \setminus A$. Since $r(U \setminus A) = r$, this maximal independent set is a base of $\mathcal{I}$, and by construction $B^* \subseteq U \setminus A$, so $B^* \cap A = \emptyset$.

Now, assume for the sake of contradiction that for all $x \in B \setminus A$, $A \cup \{x\} \notin \mathcal{I}'$. This implies that no element in $B \setminus A$ lies outside of B^*, i.e., $B \setminus A \subseteq B^*$.

Since $I = B_B \setminus A \subseteq U \setminus B$, I contains no elements from B. Since $B \setminus A \subseteq B^*$ and $B^* \subseteq U \setminus A$, all elements of $B \setminus A$ lie in $B^* \setminus I$, so:

$$|B \setminus A| \leq |B^*| - |I| = r - |B_B \setminus A|$$

Since $B_B \subseteq U \setminus B$, we have $B_B \cap B = \emptyset$, so $B_B \cap A = B_B \cap (A \setminus B)$, which gives $|B_B \cap A| \leq |A \setminus B|$. Therefore:

$$|B \setminus A| \leq r - |B_B \setminus A| = r - (|B_B| - |B_B \cap A|) = |B_B \cap A| \leq |A \setminus B|$$

Using the fact that $|X \setminus Y| = |X| - |X \cap Y|$, we get:

$$|B| - |A \cap B| \leq |A| - |A \cap B| \implies |B| \leq |A|$$

This directly contradicts our initial assumption that $|A| < |B|$.

Therefore, our assumption must be false. There must exist at least one $x \in B \setminus A$ such that $x \notin B^*$. Since $x \in B \setminus A$, we have $x \notin A$. Combined with $x \notin B^*$, we get $B^* \cap (A \cup \{x\}) = \emptyset$, i.e., $B^* \subseteq U \setminus (A \cup \{x\})$, which implies $A \cup \{x\} \in \mathcal{I}'$.

□